

CHAPTER SIX

ADVANCE TOPICS

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Chapter 6:Advance Topics

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Multiprocessing System

- Multiprocessing System consist of two or more processors of comparable capability that can share common resources such as memory, I/O devices etc., improving overall performance of system
- These processor shares the resources either through same channels that provides path to the same devices
- The system is controlled by integrated operating system that provides interaction between processor and their programs
- The organization of multiprocessor system can be divided into following types:
 - Time Shared or Common Bus
 - Multiport Memory
 - Central Control Unit

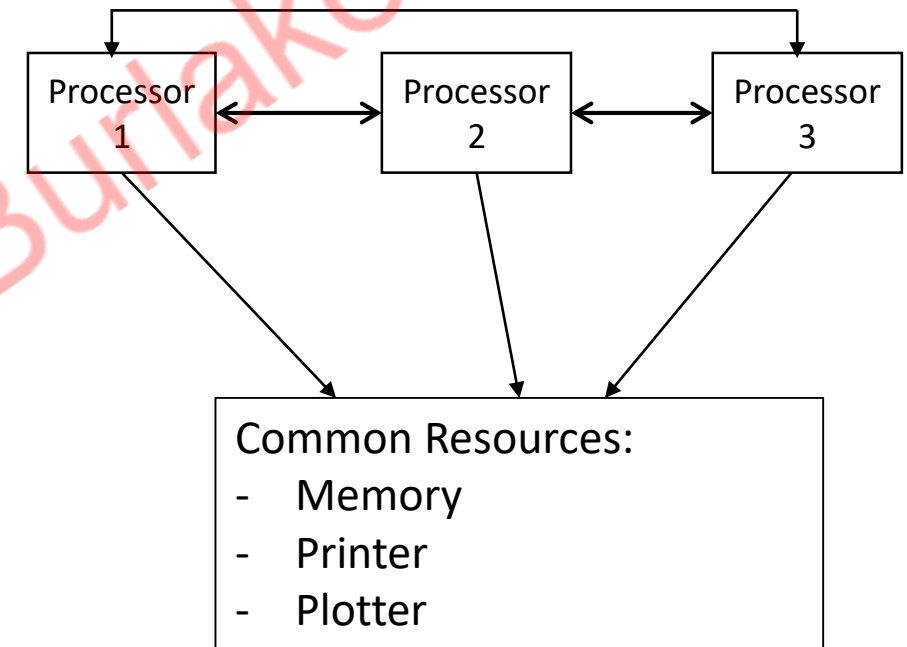


Fig: Organization of Multiprocessing system

1. Time Shared or Common Bus

- It is simplex mechanism in which the no. of CPUs, I/O modules, Memory modules are connected to the same bus
- This system must distinguish between the module on the bus to determine the source and destination of data
- When one is controlling or using bus, other device should be locked out.
- Further access to each device is divided on the basis of time

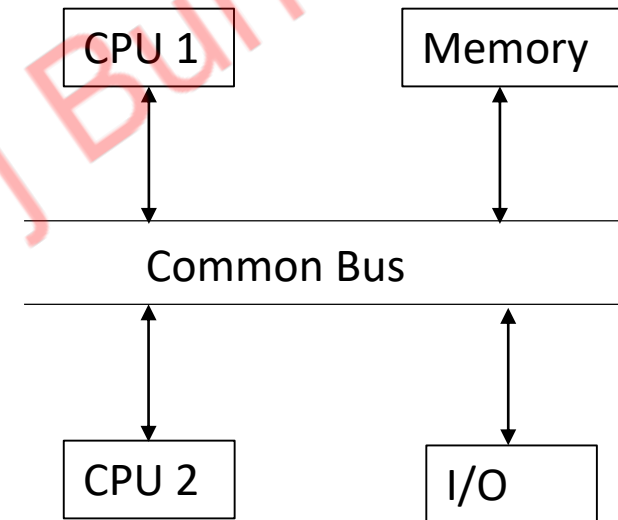


Fig: Time Shared System

1. Time Shared or Common Bus

- Advantages:
 - Simplicity: Simple, Very much similar to uniprocessor system
 - Reliability: Failure of any one system does not cause failure in whole system
 - Flexibility: Easy to expand the system by attaching more processor
- Disadvantage:
 - Performance of whole system is limited by the common bus performance

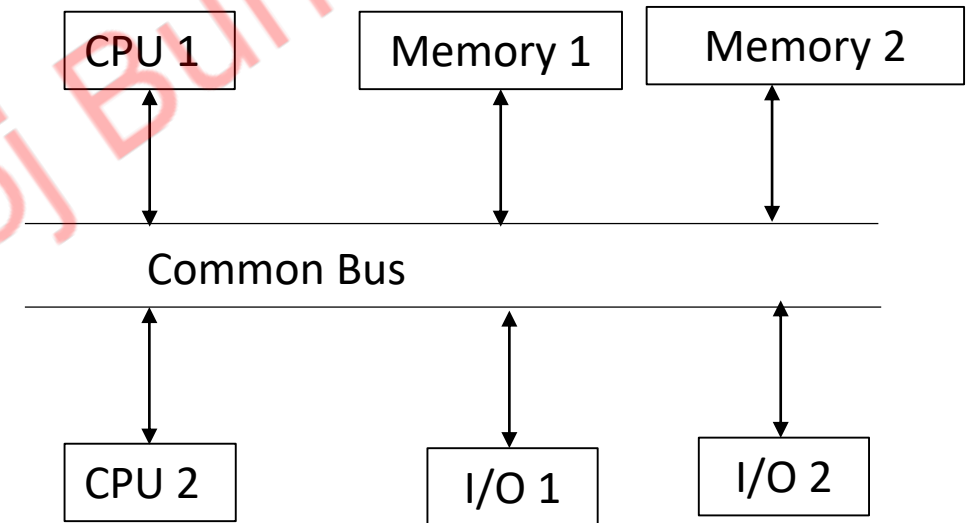


Fig: Time Shared System

2. Multiport Memory

- Multiport memory allows direct and independent access of memory modules by each CPU
- Each processor have dedicated path to each memory module with a common memory to communicate between the two processors
- This gives better performance but complex than shared bus system
- The mechanism for resolving conflicts while using memory modules should be implemented
- For that either permanently designated priorities for each module should be defined or certain portion of memory can be configured as private to one of the CPU.

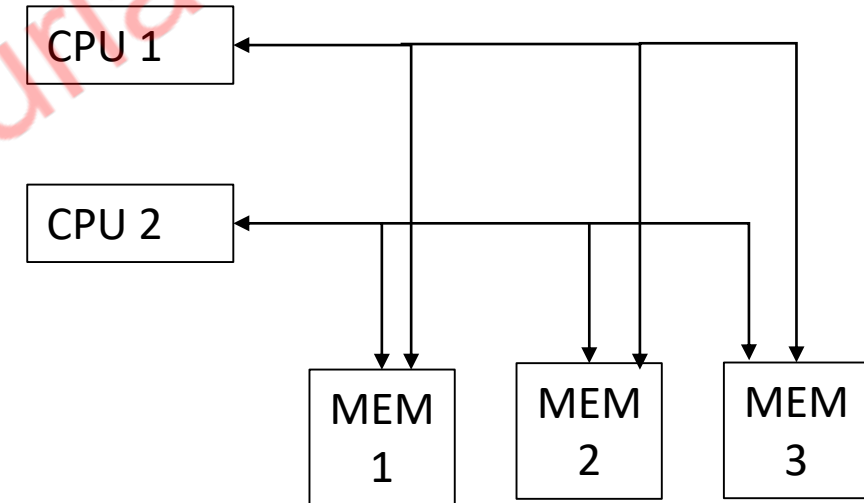


Fig: Multiport Memory System

3. Central Control Unit (CCU)

- The centralized control unit manages the transfer of data streams back and forth between independent modules by CPU and I/O, Memory.
- The controller can manages the timing function and also pass control and status signals between processors
- Since logic for coordinating the multiprocessor configuration is centered in CCU.
- The interface from I/O, memories and CPU remains undisturbed
- So interfacing is simple but designing CCU is complex

Parallelism

- It is a concept used to provide simultaneous data processing task for the purpose of increasing the computational speed of computer system
- **Pseudo Parallelism**
 - The time of processor is divided among the processes and rapid switching takes place between them
 - So it is not exactly a parallelism but gives sense of parallelism, so called pseudo parallelism
- **Real Parallelism**
 - Multiple processor executes multiple processes at a time

Parallelism

Pseudo Parallelism	Real Parallelism
1. Multiprogramming within same processor	1. Multiprogramming on different processor
2. With sharing resources (h/w & s/w)	2. Separate resources (also called H/W parallelism)
3. Overlap individual machine operation, so that they execute in parallel	3. Having separate processors getting separate chunks of program

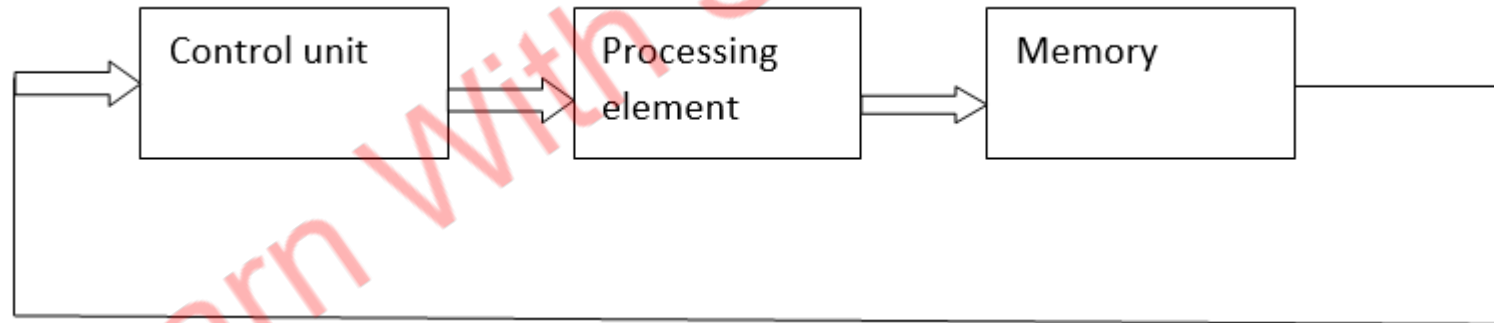
Flynn's Classification

- Based on the no. of instruction and data items that are manipulated simultaneously
 - Single Instruction, Single Data (SISD) Stream
 - Single Instruction, Multiple Data (SIMD) Stream
 - Multiple Instruction, Single Data (MISD) Stream
 - Multiple Instruction, Multiple Data (MIMD) Stream

		Instruction Streams	
		one	many
Data Streams	one	SISD traditional von Neumann single CPU computer	MISD May be pipelined Computers
	many	SIMD Vector processors fine grained data Parallel computers	MIMD Multi computers Multiprocessors

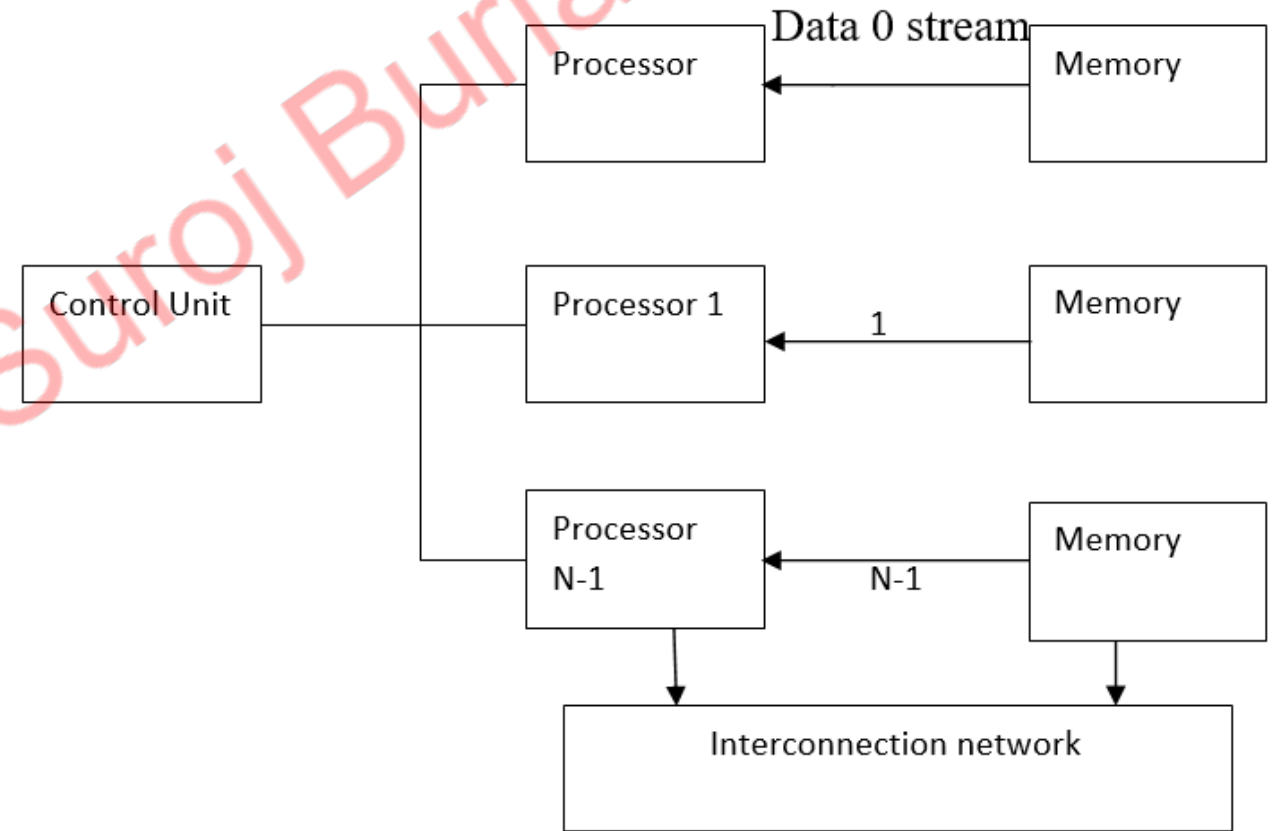
1. Single Instruction, Single Data (SISD) Stream

- Here, the single processor executes a single instruction to operate on data stored in single memory
- E.g. Uniprocessor



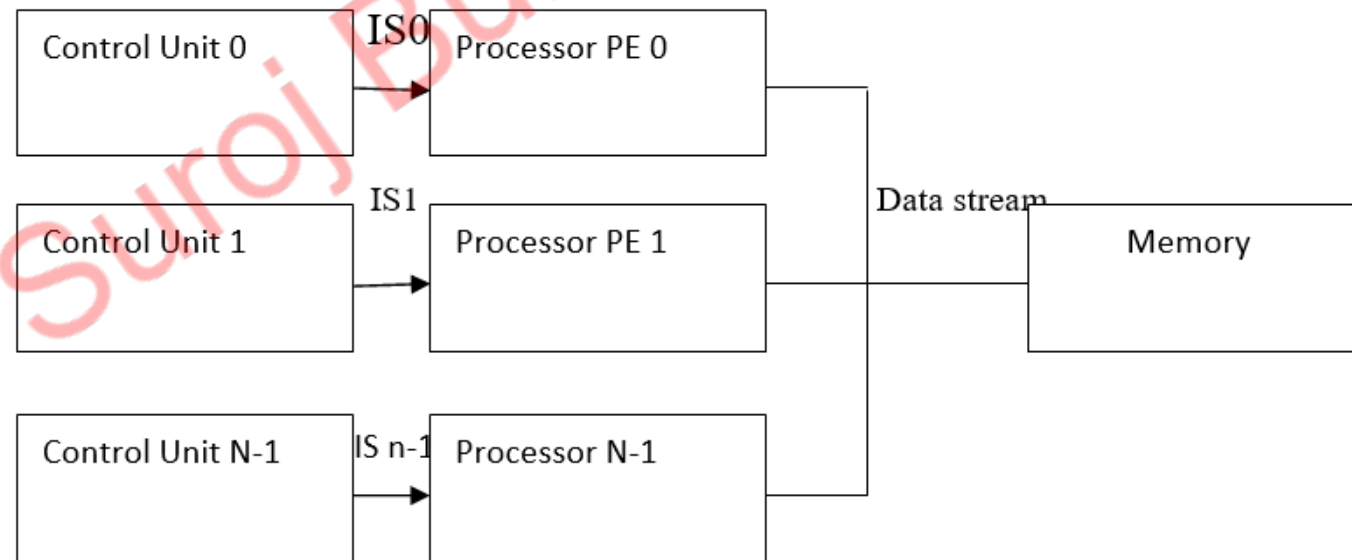
2. Single Instruction, Multiple Data (SIMD) Stream

- Single machine instruction controls the simultaneous executing of no. of data items.
- i.e. the processor receives a same instruction from single control unit and then those operates on different set of data as available to them
- E.g. Vector Processor, Array Processor



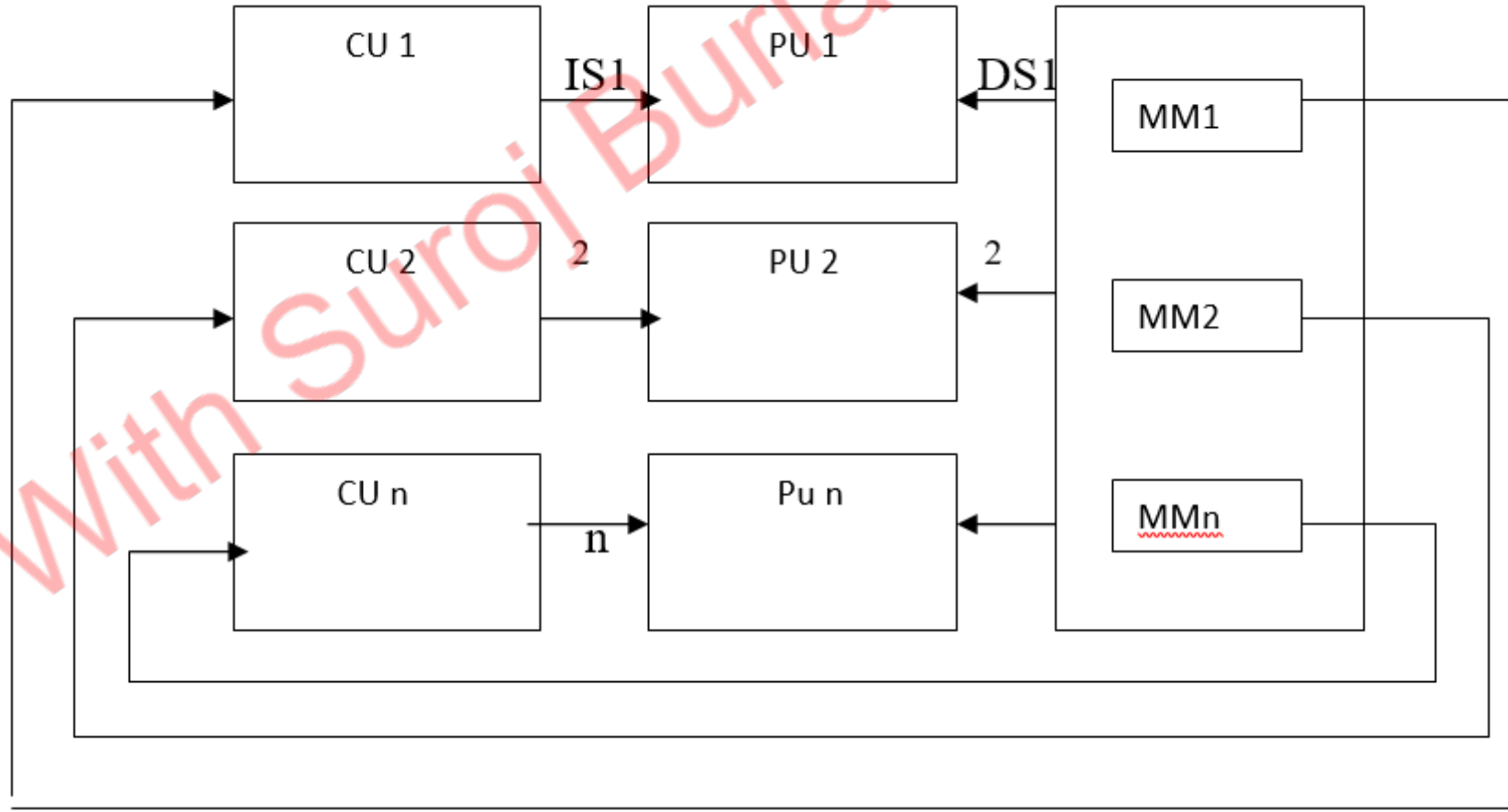
3. Multiple Instruction, Single Data (MISD) Stream

- Single sequence of data is transmitted to a set of processor simultaneously, each of which execute different instructions in sequence
- No practical system constructed yet, only theoretical
- May be a pipelined computers



4. Multiple Instruction, Multiple Data (MIMD) Stream

- A set of processor simultaneously execute different instruction sequences on different datasets
- E.g. Multiprocessor



Instruction Level, Thread Level and Process Level Parallelism

- Process:
 - It is an instance of program loaded in memory for execution
- Thread:
 - Small chunks of process

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Instruction Level Parallelism

- It is a degree to which the instruction of program can be executed in parallel
- Typically multiple instructions are faced at a time and then executed in parallel if both are independent

Process Level Parallelism

- It is a degree to which the process of program are loaded in memory parallel
- System with multiple process will yield better performance than one with single processor of same
- Concerned with scheduling, execution and resource
- It is a bit time consuming

Thread Level Parallelism

- The process is divided into several smaller streams called threads such that they can be executed in parallel
- Concerned with scheduling and execution
- Thread switching consumes less time than process switching

Data Parallelism

- Data Parallelism is parallelism inherent in program loops, which focuses on distributing a data across different computing nodes to be processed in parallel

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Interprocess communication

- In any system, multiple processes may be active in one time
- The process of communication between those multiple process is called interprocess communication
- It deals with the how one process can pass information to another thus making sure that processes do not get into each other's way
- Interprocess methods are message passing, shared memory and remote procedure calls
- The several reasons for providing an environment that allows process cooperation are information sharing, speed up, modularity, convenience, privilege separation etc.
- Also referred as inter-thread communication or inter-application communication

Resource Allocation

- Resource allocation plays vital role in case of multiprocessing systems or system with multiple processes because the multiple processes/processors might need to share same communication resources
- If the resources are not located efficiently, the use of multiple processes/processors only won't enhance the overall performance of system
- Resources could be computer's memory, data in device, interface buffer, one or more files or the required amount of processing power

Deadlock

- The responsibility of operating system to see that resources are used efficiently, otherwise the use of multiple process or processors could be resulting into a system deadlock that will freeze any process happening currently
- Conditions for deadlock:
 - Mutual Exclusion: shared resources are used in a mutually exclusive manner
 - Hold and Wait: processor hold into resources they already have while waiting for allocation of other resource
 - No Preemption: resources cannot be preempted until the process release them
 - Circular wait: circular chain of process exist in which each process holds resources wanted by next process in chain
- Deadlock may be
 - System deadlock
 - Process deadlock

Semaphore

- Semaphore is a software technique to manage the resources
- Basically, it is a flag associated with each shared resources such that all processor or processes can access the flag and modify it
- But before accessing any resources, they will check the semaphore
- If it is '1' the resource is being used else it is available to any requesting processor

Operating System

- Operating system is a program that controls the execution of application program and acts as an interface between user and hardware of computer
- E.g. Windows, Linux, Android, IOS, etc.
- The operating system is responsible for preventing conflicting use of shared resources by several processes allocation and proper scheduling
- The basic feature of OS are:
 - Resources Allocation and Management
 - Table and Data Protection
 - I/O management
 - Memory Management
 - Abnormal Program Termination
 - Precision of System Deadlock
 - Process Scheduling
 - Scheduling for multiprocessors
 - Reconfiguration for multiprocessors

Different Microprocessor Architecture

- Register Based and Accumulator Based Microprocessor
- RISC and CISC Architecture
- Digital Signal Processor

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Register Based and Accumulator Based Microprocessor

Accumulator Based Microprocessor	Register Based Microprocessor
1. Accumulator is most significant register	1. Accumulator and other registers are identical
2. In arithmetic and logic operation one operand is always accumulator	2. Arithmetic and logic operation can be performed in any registers except I/O
3. I/O data are performed via accumulator	3. I/O operation are limited to accumulator also
4. Accumulator is directly connected to ALU while other registers are connected to ALU through temporary register or accumulator e.g. $\text{ADD R ; } A \leftarrow A + R$	4. All registers are connected to the same point to the ALU e.g. $\text{ADD R1,R2 ; } \leftarrow R1 + R2$
5. e.g. 8085 is accumulator based microprocessor	5. e.g. 8086 is register based microprocessor

RISC and CISC Architecture

- CISC Based Architecture
 - Stands for Complex Instruction Set Computer
 - Characteristics of CISC computer are:
 - Large number of complex instruction
 - Requires multiple cycle for their execution
 - Less registers, so frequent memory access, so reduced processing speed
 - Variable length instruction format is used
 - Instruction implemented in microprocessor and interpreted by microprogram
 - Large variety of addressing modes
 - Less pipelining used
 - Low cost
 - For general purpose application, where speed is not so significant
 - E.g. intel 3090, 8086, 80286, Pentium, AMD etc

RISC and CISC Architecture

- RISC Based Architecture

- Stands for Reduced Instruction Set Computer
- Characteristics of RISC computer are:
 - Less number of simple instruction
 - Instructions are short and of equal length
 - One cycle per execution
 - Less addressing mode
 - More registers, so less memory reference
 - Memory reference limited to Load/Store
 - Instruction implemented in hardware, no microprogram
 - Heavy pipelining
 - Used for overlapped register window
 - Used in scientific and research oriented task where high speed is required
 - E.g. DEC Alpha, Power PC, ULTRA SPARC etc.

Digital Signal Processor (DSP)

- DSP is a special purpose CPU that provides ultra fast instruction sequences, such as shift and add, multiply and add, which are commonly used in math intensive signal processing applications
- The digital signal obtained from ADC can be operated upon by adding extra features to them, reducing redundancy of the signals, making them immune to noise etc. by some external processor called digital signal processor
- Various DSP functions can be implemented using hardware and software. The software implementation is slow and hardware implementation require different architecture than von newman
- DSP architecture is based on havard architecture where separate space for program and data
- They are targeted for some specific application such as Sound Card, Video Card, Fast Fourier Transform (FTT), etc.
- Specialize addressing modes
- Single cycle – multiply accumulate capability
- E.g. Texas instrument, TM S320C, Motorola 3600

Digital Signal Processor (DSP)

- Advantages

- Immune to noise
- More accurate than analog
- Versatile, flexible
- Provides various configuration of on-chip memory

- Disadvantages

- Requires high bandwidth
- Depends upon other device as analog to digital conversion and vice versa required
- complexity



Thank You

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